

Cell:- 14

Yes. Let's ignore the individual Python lines and look at **what Cell 14 actually means architecturally**.

Cell 14 — Actual Role in the Retriever

Cell 14 is the preparation and validation stage of your GraphRAG retriever.

It takes the **RDF knowledge graph that you already built** and prepares the information needed for the retriever to later answer a user query.

It does **not retrieve anything yet**.

The logic is:

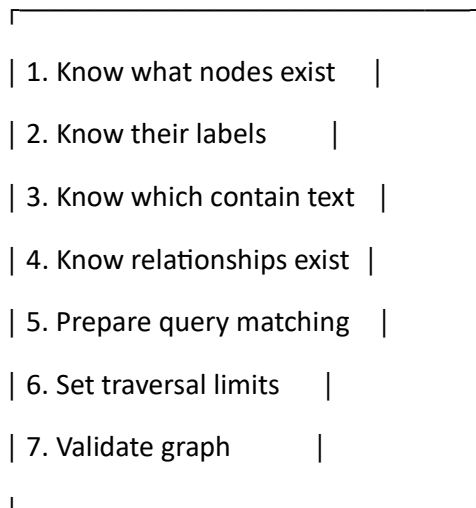
Existing RDF Knowledge Graph



Cell 14



Prepare Retriever



Retriever is ready



Cell 15+

Think about it like this

Suppose your Knowledge Graph contains:

Engine Control

|

| depends_on



Hydraulic System

|

| connected_to



Pump System

and each document node also has its actual XML/document content.

Now the user asks:

"What does the engine control depend on?"

Before you can answer this question, your retriever needs to know:

1. **What entities/documents are in the graph?**
2. **What are their human-readable names?**
3. **Which graph nodes have actual document content?**
4. **What relationships connect those nodes?**
5. **How do I match "engine control" to the correct graph node?**
6. **How far should I traverse the graph?**
7. **How much information should I retrieve?**

Cell 14 prepares all of this.

The 4 Main Jobs of Cell 14

Actually, you can explain the entire cell in **four major responsibilities**.

1. Prepare the searchable nodes

It creates an index of:

Label → Graph Node

For example:

"engine control"



doc_101

This is important because the user gives a **text query**, while the graph contains **RDF nodes**.

So Cell 14 prepares the bridge:

User's words



Graph node

2. Identify the nodes that contain actual knowledge

It finds which nodes have:

documentContent

This is important because finding a graph node alone isn't enough.

You eventually need:

Graph Node

+

Document Content

so that the LLM can receive actual source information.

Therefore:

Node identification



Content availability



Grounded retrieval

3. Prepare the graph traversal

Your retriever is not supposed to search the whole graph.

Cell 14 defines rules such as:

Start from relevant nodes



Explore up to 2 hops



Reduce importance of farther nodes



Limit number of nodes/edges

For example:

2-hop



Engine Control → Hydraulic → Pump



1-hop

The retriever can therefore use the **structure of the knowledge graph** rather than just searching text.

This is the key difference between your GraphRAG retriever and a normal vector retriever.

4. Verify that the graph is actually usable

This is the **validation/gate** part.

Cell 14 checks:

Do nodes have labels?



Do nodes have document content?



Do graph relationships exist?

If something is missing:

Broken / incomplete graph



STOP

Instead of allowing the retriever to continue and potentially give the LLM an answer with no real evidence.

Why is Cell 14 needed?

Imagine you didn't have Cell 14.

Cell 15 receives:

"What does Engine Control depend on?"

and immediately tries to retrieve.

But it doesn't know:

Where are the nodes?

Which node represents Engine Control?

Which nodes have content?

Are relationships available?

How many hops should I search?

How much context should I return?

So Cell 14 establishes all those things **before the first query comes in**.

Very Important: Cell 14 is NOT Retrieval

This distinction is important when you explain your project.

Cell 14

Preparation

+

Index creation

+

Configuration

+

Validation

Cell 15 onward

User Query



Query understanding/matching



Seed node selection



Graph traversal



Candidate collection



Ranking



Document/context retrieval

So you should **not say**:

"Cell 14 retrieves the relevant documents."

Instead say:

"Cell 14 prepares and validates the graph-based retrieval environment. It builds the node-label index, identifies nodes containing document content, verifies that graph relationships are available for traversal, and configures the limits and matching strategy that the downstream retriever will use."

Why did we choose this approach?

Because your graph contains **20K+ XML documents**.

You cannot efficiently do:

User Query



Search every XML



Search every relationship



Send everything to LLM

Instead, you prepare indexes and constraints:

20K+ XML documents



Knowledge Graph



Cell 14 prepares indexes



Query only relevant nodes



Traverse relevant graph area



Retrieve limited context



LLM

This makes the retrieval process:

- **more targeted**
- **more scalable**
- **more grounded**
- **less noisy**
- **less expensive**

The simplest way to understand Cell 14

Think of Cell 14 as **setting up a library before searching for a book**.

You don't ask:

"Where is the book?"

before you know:

- what books exist,
- where they are indexed,
- which books have full content,
- how the shelves are connected,
- and how much you are going to search.

Cell 14 does that preparation for your **Knowledge Graph**.

One-line explanation

Cell 14 prepares the Knowledge Graph for retrieval by creating the searchable node index, identifying content-bearing nodes, validating graph relationships, and setting the rules for how much and how far the retriever should search.

That is the **actual role of Cell 14**.

explain me with one example of cell 14

Sure. Let's take **one realistic example** and see exactly what Cell 14 is doing.

Example

Assume your Knowledge Graph has these three documents:

Document A: Engine Control

Document B: Hydraulic System

Document C: Hydraulic Pump

And the relationships are:

Engine Control

|

| depends_on



Hydraulic System

|

| contains



Hydraulic Pump

Each document node also has its actual document content.

Now, before the user asks anything, **Cell 14 runs**.

What Cell 14 does with this graph

Step 1 — It identifies the searchable entities

Cell 14 looks at the graph labels and creates something conceptually like:

"engine control" → Document A

"hydraulic system" → Document B

"hydraulic pump" → Document C

This is important because later the user might ask:

"What does the engine control depend on?"

The retriever needs to convert the words "**engine control**" into the actual graph node **Document A**.

So Cell 14 prepares that mapping.

Step 2 — It identifies which nodes have actual content

It also checks:

Document A → has document content ✓

Document B → has document content ✓

Document C → has document content ✓

Why?

Because finding a node is not enough.

Eventually, the LLM needs the **actual information from the document**, not just the node name.

So Cell 14 knows:

"These are the nodes from which I can actually retrieve source content."

Step 3 — It checks whether relationships are available

It verifies that the graph actually contains relationships such as:

A —depends_on—> B

B —contains—> C

If these relationships don't exist, graph traversal cannot work.

So Cell 14 essentially asks:

"Can I actually walk through this graph?"

If the answer is no, it stops the pipeline.

Step 4 — It prepares the traversal rules

Cell 14 says:

Start from a relevant node

↓

Search up to 2 hops

↓

Don't explore too many nodes

↓

Don't collect too many relationships

So for our example:

Engine Control

↓ 1 hop

Hydraulic System

↓ 2 hop

Hydraulic Pump

The retriever is allowed to reach the pump, but it doesn't continue endlessly through the whole KG.

Step 5 — It prepares fuzzy matching

Suppose the graph label is:

"Engine Control System"

but the user asks:

"Tell me about engine control"

Cell 14 has prepared fuzzy matching so that the retriever can recognize:

"engine control"



"engine control system"



Document A

So the user doesn't have to type the **exact RDF label**.

Then what happens?

After Cell 14 finishes, your retriever is **ready**.

The next cell can receive:

"What does the engine control depend on?"

and then actually perform retrieval:

User Query



"engine control"



Find matching node



Engine Control (Document A)

|

| graph traversal



Hydraulic System (Document B)

|



Retrieve document content

|



Relevant context

|



LLM

So what is Cell 14 actually doing?

Think of it as **preparing the map before starting the journey.**

Your KG is the map:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow \dots$

Cell 14 prepares the retriever by saying:

"Here are the places I can search, here are the names of those places, here are the places that contain useful information, here are the roads connecting them, and here are the rules for how far I am allowed to travel."

It doesn't travel yet.

The actual travel/search happens in the next retriever cells.

In one sentence for your project explanation:

Cell 14 prepares and validates the Knowledge Graph for retrieval by indexing the graph nodes and their labels, identifying nodes containing document content, verifying that relationships are available for traversal, and defining the limits and matching rules that the actual retriever will use.

That is the correct conceptual role of **Cell 14**.

